

Applying Radio Environment Maps to Cognitive Wireless Regional Area Networks

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Abstract—The IEEE 802.22 wireless regional area network (WRAN) is the first worldwide commercial application of cognitive radio (CR) networks refarming the TV broadcast bands. According to US FCC's recent public notice, WRAN products are scheduled to be available for the market by February, 2009. This paper first presents a brief review of the IEEE 802.22 WRAN standardization, and then introduces the radio environment map (REM) as an innovative cost-efficient approach to developing and managing WRAN systems. REMs can provide powerful infrastructure support to the functionality of the WRAN cognitive engine (CE). The data model of the REM is presented together with an extensive discussion on how to exploit the REM for a variety of applications in WRAN systems, focusing on the REM-enabled case- and knowledge-based learning algorithms (REM-CKL) for WRAN CEs. Furthermore, REM-based radio scenario-driven testing (REM-SDT) is also presented as a viable approach to evaluating the performance of CEs. Future research topics are discussed in the final section.

Index Terms—case- and knowledge-based learning, cognitive radio, IEEE 802.22 WRAN, radio environment map (REM)

I. INTRODUCTION

On December 19, 2005, the House of Representatives of the United States approved legislation to complete the country's transition to new, higher-quality digital television (TV) by February 17, 2009. Refarming the analog TV spectrum promotes new wireless communication technology developments, most notably the cognitive radio (CR) technologies. IEEE 802.22 will be the first worldwide CR-based standard to support the unlicensed operation in TV bands (54-862 MHz), which is to coexist with incumbent users and provide wideband internet access to rural and suburban areas. According to US FCC's recent public notice, products refarming TV bands are scheduled to be available to the market by February, 2009 [1]. The IEEE 802.22 Work Group kicked off in November 2004 and approved the functional requirement document for WRAN systems in September 2005 [2]. Ten initial proposals were merged into a single one in March 2006, and the draft standard (D0.1) was developed in May 2006 [10]. The complete 802.22 standard is expected to be approved by May 2007. For 802.22 WRAN systems, the primary users (PU), those with priority rights, mainly include incumbent analog and digital TV stations, TV translators, TV boosters, TV receivers, and wireless microphones.

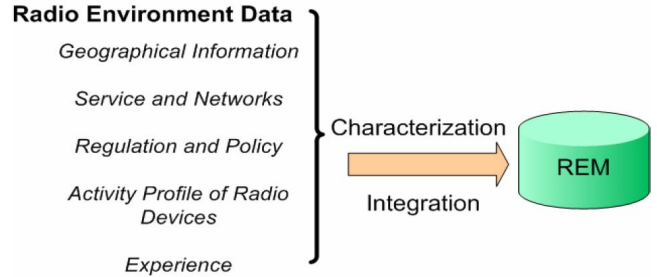


Fig. 1. REM is envisioned as the navigator of CRs, by characterizing the radio scenario and offering network support and prior knowledge

WRAN systems, including base stations (BS) and customer premise equipment (CPE), are secondary users (SU), and therefore, should avoid generating harmful interference to the PUs.

A map is the most efficient way to describe the real world. The radio environment map (REM) is proposed as an integrated database that characterizes the radio environment for CRs. The REM, maintained by distributed CR nodes and/or network infrastructure, is envisioned as the navigator for CRs. As illustrated in Fig. 1, REM covers multi-domain environmental information, such as geographical features, available services, spectral regulations, locations and activities of radios, relevant policies and past experiences. The REM information can be updated with observations from CR nodes and disseminated throughout CR networks [3][4][5]. The REM can also be viewed as an extension to the available resource map (ARM) proposed by Krenik for CR applications in unlicensed wide area networks [6][7]. The motivation of this paper is to explore the potentials of REM in cognitive WRANs.

The remainder of this paper is organized as follows: section II presents a reference data model for the REM at WRAN BSs; section III overviews nine potential applications of REMs to WRANs; section IV introduces a REM-CKL algorithm with some preliminary simulation results; section V concludes this paper with some insights and future research topics.

II. REM DATA MODEL FOR COGNITIVE WRAN SYSTEMS

A. The role of databases in CR networks

If we envision CRs' activities as "radio transactions," then it becomes apparent that databases would be at the heart of

future CR networks for various cognition functions, such as situation awareness, case-based learning (CBL) and knowledge-based learning (KBL). This motivates us to develop an integrated database, REM, which combines environmental information, past experience and radio knowledge altogether. The goal is to enable CR to become situation-aware by simply referencing the REM. In this way, the processing power and adaptation time of CR could be significantly reduced. Low-cost large-scale CRs also become possible if they could access, share and exploit the local and/or global REMs [3]. Note that database technologies have been developing rapidly to meet the challenging requirements of new applications, such as web-based and mobile databases [8].

B. REM data model for cognitive WRAN BS

As shown in Fig. 2, the cognitive WRAN BS could consist of a spectrum sensing module, REM, CE, and (software-defined) radio transceiver. The CE may have case- and knowledge-based learning (CKL) modules, a channel modeler and predictor, a multi-objective optimizer, a spectrum manager and other functional entities. The CE first obtains situation awareness by querying the REM, and then determines the utility function that best fits its current situation [9]. Note that the multi-objective optimizer helps the CE to reach multiple goals, such as minimizing harmful interference and maximizing spectrum efficiency. Fig. 3 shows a reference entity-relationship (E-R) model for REM design. The WRAN BS serves a number of CPEs, co-exists with incumbent PUs and other neighboring WRANs, and also complies with local regulations and policy.

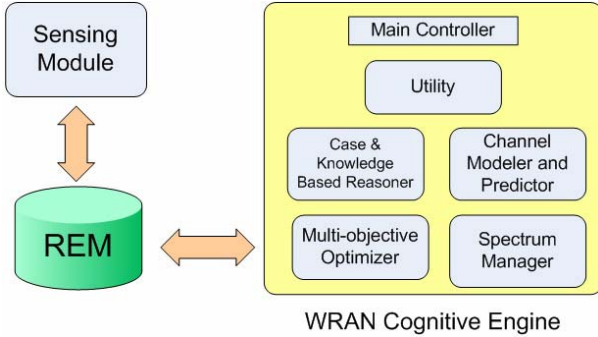


Fig. 2. Architectural diagram of REM-based CE for 802.22 WRAN BS

C. APIs between REM and CE

The REM interfaces with the CE by common application program interfaces (APIs). For a CE to be useful across a variety of different environments it must operate at a level of relative independence from the environment. By defining proper APIs between REM and CE, the REM and CE can be developed and/or updated independently of each other.

III. APPLICATIONS OF REM TO COGNITIVE WRAN

Compared to traditional fixed wireless access systems, the most distinctive features of cognitive WRANs are the awareness and protection of PUs, as well as the autonomous optimization and management of radio resources. Potential applications of REM in cognitive WRAN systems may

include, but are not limited to, the following:

(i) Network initialization: “The WRAN BS starts by consulting the TV usage database and the regional WRAN information base to find potentially empty channels,” [10]. The BS can be initialized by downloading relevant REM entries.

(ii) Efficient spectrum sensing and optimal channel assignment: the most important functionality of WRAN BS is to allocate radio resource to CPEs such that it minimizes harmful interference to PUs. From historical REM information, the BS can derive the usage patterns of both PUs and SUs, which can be exploited for efficient spectrum sensing and channel state prediction [9].

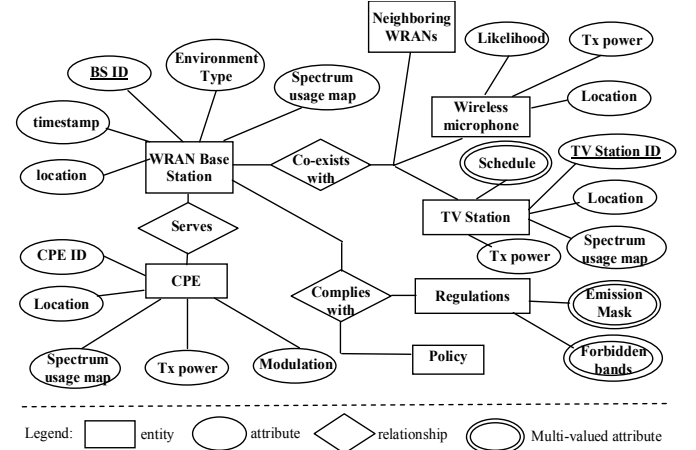


Fig. 3. Entity-Relationship (E-R) diagrams of REM for WRAN BS

(iii) CPE transmission power control: based on the selected operation channel and the REM, the allowed maximum transmission power of each CPE can be determined with the procedure defined by the 802.22 draft standard [10].

(iv) Awareness and protection of PUs: by exploiting REM-CKL algorithm (to be explained in section V), typical hidden PU scenarios could be identified or anticipated promptly by learning from previous experience. Therefore, pre-emptive measures could be taken to avoid or mitigate the interference subjected to the hidden PU node. For example, WRAN CE can anticipate the presence of (hidden) PU nodes for certain radio scenarios. Statistical analysis can be employed to derive usage patterns, such as duty ratios, arrival rates, and intervals of arrival. TV channel statistics, such as channel “reputation,” which indicates the probability of presence of PUs at a TV channel (as shown in Fig. 4), can be obtained by analyzing the historical radio environment information. TV channel reputation can be used for ranking the candidate channels. In this way, the possibility of interfering with PUs could be further reduced. Prior knowledge of the radio environment, such as spatiotemporal statistics of the PUs, can also help a CR to improve the PU detection rate by adjusting the detection threshold [9]. Furthermore, by Monte Carlo simulation, with location information of radio nodes, the WRAN CE may predict the possible areas where PUs might not be detected and then take pre-emptive measures accordingly.

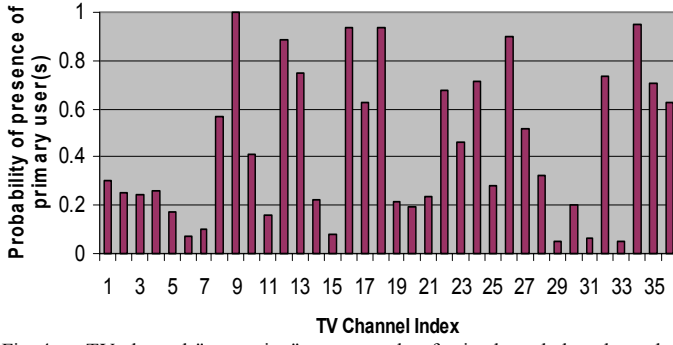


Fig. 4. TV channel "reputation", an example of prior knowledge about the radio environment obtained through statistical analysis of REMs

(v) Fast adaptation: by leveraging prior knowledge of the radio environment and applying REM-CKL techniques, the BS can become immediately aware of its current radio scenario to narrow down the solution space over which it must search. Therefore, it can make faster adaptations and evacuate the channel more quickly when PUs re-appear.

(vi) Radio resource management and optimization: during different phases of WRAN network deployment or during different times of the day, the WRAN service providers may want to have different goals or priorities. For example, at the initial stage the operator usually has higher priority to the network coverage. However, when more CPEs need to be served, e.g., during the peak hours of the day, improving the network throughput and spectrum efficiency becomes a major concern. Ideally, upon service request from CPE(s) and REM information, the WRAN CE could adaptively determine an appropriate utility function to fit its current situation, rather than using a static utility function.

(vii) REM-based CPE positioning enhancement: by taking advantage of CPEs' wideband spectrum sensing capabilities, wideband RF fingerprinting techniques can be exploited to enhance the CPE ranging. For example, triangulation-based location methods can be employed with more reference radio stations from various wireless systems other than WRANs. The advantage of this method is more apparent when the number of WRAN BSs is less than the number required for traditional location techniques.

(viii) REM-enabled WRAN interference management and network security: a detection scheme for malicious PU emulation attacks could be based on location verification, as proposed in [11]. Furthermore, such attacks could also be detected based on the unique features of PU mobility, usage patterns, and transmission power. PU detection, classification and location can be conducted at the BS with the help of the REM to further improve interference management and security of WRAN systems. For example, WRAN systems can set up authorized service areas in which only registered CPEs can access the network.

(ix) Performance evaluation and testing of CRs: testing CRs is much more challenging as they operate very differently from traditional radios, due to their flexibility, learning capabilities, and the demanding or unpredictable operation environments. The most accurate predictor of future performance of CR is to emulate it in a similar situation (not

unlike the behavior-based interview). REM-based radio scenario-driven testing (REM-SDT) is a viable approach to evaluating the performance of the WRAN CE.

REM could also be used as a virtual "radio environment generator" together with other test equipment, such as arbitrary waveform generators, for testing CRs under various radio scenarios. The CR under test is subjected to various realistic situations stored in the REM which could be in form of machine-readable XML files. For example, the CR tester could emulate various PU waveforms with certain usage patterns in certain frequency bands, and then measure the performance of the CR under test through its RF emission. This can indicate the cognition levels and the effects of its adaptations. For WRAN systems, typical radio scenarios, extreme scenarios (e.g., no CPE is active in the BS service area), and hidden PU scenarios should be considered when testing WRAN CEs.

IV. REM-ENABLED CASE AND KNOWLEDGE BASED LEARNING

CBL is an inductive learning technique, which recalls previous experiences to solve current problems [12][13]. KBL is a deductive learning technique that employs prior knowledge for similar purposes. REM-CKL is proposed to leverage both CBL and KBL techniques in conjunction with the REM.

A. Motivations of REM-CKL for cognitive WRANs

WRAN systems have the following unique features as compared to cellular systems: the radio links of WRAN systems are quasi-static, considering that both PU and SU nodes could be assumed to be stationary; the spectrum usage patterns of PUs and SUs are usually periodic over one day and/or one week; the nature of TV signals is not bursty, and TV signals usually have predefined schedules and operate on fixed, pre-allocated TV channels and are not changing dynamically. Wireless microphone operational bands are also fixed and known. All these features motivate us to employ the CKL in WRAN CE. REM provides an efficient tool to characterize radio scenarios, which is a prerequisite for employing CKL. That is why we call it REM-CKL.

B. How REM-CKL works

A system approach that formalizes the REM-CKL is discussed as follows: first, the CR needs to be aware of its situation by characterizing and indexing the current radio scenario, and then retrieving the case library and/or knowledge base for applicable experiences and/or rules. For WRAN systems, the radio scenario can be identified by the combination of active CPE identities, the requested services, active/candidate/occupied channel sets, which altogether present the geographical layout of radio links and the required and the available radio resources at the BS.

An important step for the REM-CKL is to determine the similarity between radio scenarios. An approach to accomplish this is to define the similarity function as follows:

$$f(\cdot) = w_1 * \Delta_1 + w_2 * \Delta_2 + w_3 * \Delta_3$$

where w_1 , w_2 and w_3 are weights to the "radio link similarity," "radio environment similarity," and "radio resource

similarity,” respectively; Δ_1 , Δ_2 and Δ_3 represent the similarity of geographical layout of CPEs and their service requests, the similarity of RF environment, and the similarity of available radio resource at WRAN BS, respectively.

REM-CKL can provide a good starting point for CRs and help accelerate the optimization process. For WRAN applications, after the current radio scenario is characterized, KBL is applied with the current domain knowledge to seek an appropriate solution. If none can be found, the CE searches its previous experience through the CBL. Emulations and/or field testing can be employed to initialize the case library, which should be updated continually with feedback information from the network operations. Note that the services and QoS requirements of 802.22 WRANs are quite similar to those of 802.16 WiMAX systems [14].

C. Preliminary simulation results

Fig. 5 shows the WRAN BS CE performance simulation results in terms of adaptation time under five different radio scenarios. For each scenario, a number of new connections are added to an existing WRAN network. Compared to genetic algorithms (GA), when using the REM-CKL algorithm, the WRAN CE makes adaptation significantly faster, which is critical for time-sensitive CR applications.

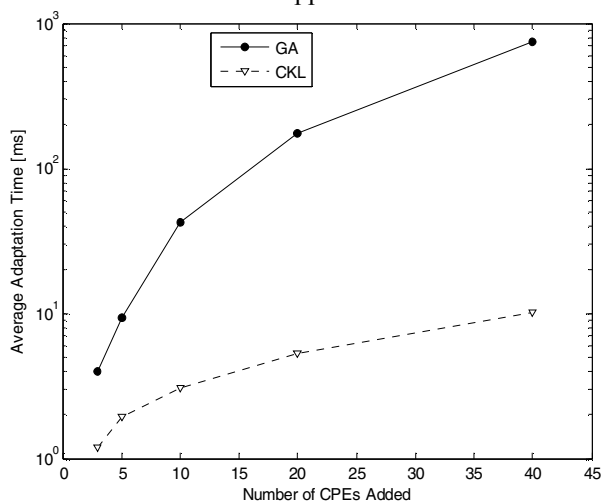


Fig. 5. Average adaptation time resulting from REM-CKL vs. GA

V. SUMMARY AND FUTURE WORK

802.22 WRAN is the first application of CR networks refarming the TV broadcast bands. The standardization and deployment of WRAN systems will have significant impact on the future advancement of CR design. The REM, being envisioned as an integrated database for CRs, offers powerful network support to cognitive WRAN systems in many ways. The key idea behind the REM is digitizing and indexing radio environment information for CR access and sharing. This paper introduces a comprehensive cost-efficient approach to developing and managing WRAN systems based on the REM. Nine potential applications of REM in WRAN systems are examined, focusing especially on REM-CKL.

Applying REMs to WRAN and beyond is an interesting inter-disciplinary research by nature. Database technology,

wireless communication and networking technology, artificial intelligence (machine learning) are all involved to fully explore the potentials of REM for CR networks. Future research topics may include the following: improve the case indexing, retrieval and adaptation algorithms for REM-CKL; evaluate WRAN CE performance under more sophisticated radio scenarios; research on mobility support for cognitive ad hoc WRANs and convergence with 802.16 WiMAX.

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